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Calibration method and calibration system for radio frequency transceiver

Abstract

A calibration method for a radio frequency (RF) transceiver and a calibration system thereof are provided. The calibration method includes configuring a controller to perform a first transmission test on the RF transceiver and generate a plurality of model test results for selecting a target calibration model from a plurality of reference calibration models to calibrate the RF transceiver; configuring the controller to perform a second transmission test on the RF transceiver and generate a plurality of parameter test results for selecting a target parameter set from a plurality of reference parameters to calibrate the RF transceiver; and configuring the controller to perform a third transmission test on the RF transceiver and generate a plurality of circuit test results for selecting a target simulation circuit from a plurality of reference simulation circuits to match the target simulation circuit to the RF transceiver.

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Background/Summary

RELATED APPLICATIONS

(1) This application claims priority to Taiwan Application Serial Number No. 112108781, filed Mar. 9, 2023, which is herein incorporated by reference.

BACKGROUND

Technical Field

(2) The present disclosure relates to the field of communication technology, and more particularly, to a calibration method and a calibration system for a radio frequency transceiver.

Description of Related Art

(3) With the increasing demand for 5th generation mobile communication system (5G), related industries have proposed the open radio access network (O-RAN), which is used to improve the flexibility of 5G architecture and provide smart management. Remote radio units (RRUs) and small base stations used in O-RAN can use the ADRV902x series chips to perform Digital Pre-Distortion (DPD) calibration. The existing calibration method is performed by antenna engineers having to manually input multiple sets of calibration parameters to the test platform, and then verify and record the error vector magnitude (EVM) results corresponding to each set of calibration parameters.

(4) The aforementioned manual operations are cumbersome and time-consuming, causing related

industries to spend additional development costs on manpower. In addition, because the existing calibration method requires multiple manual comparisons and data verifications, the performance of the calibrated RRU products is inconsistent, so that they cannot all meet the specification requirements of the 3rd generation partnership project (3GPP). In view of this, a calibration method and a calibration system that can effectively reduce manual testing man-hours and improve RRU performance are needed by the market and related industry.

SUMMARY

(5) According to one aspect of the present disclosure, a calibration method for a radio frequency transceiver includes a first calibration step, a second calibration step, and a circuit matching step. The first calibration step includes configuring a controller to execute a first calibration software module for performing a first transmission test on the radio frequency transceiver to generate a plurality of model test results so as to select a target calibration model from a plurality of reference calibration models, and to calibrate the radio frequency transceiver according to the target calibration model. The target calibration model corresponds to a lowest value among the plurality of model test results. The second calibration step includes configuring the controller to execute a second calibration software module for performing a second transmission test on the radio frequency transceiver to generate a plurality of parameter test results so as to select a target parameter set from a plurality of reference parameters, and to calibrate the radio frequency transceiver according to the target parameter set. The target parameter set corresponds to a value set of the plurality of parameter test results. The circuit matching step includes configuring the controller to execute a circuit matching software module for performing a third transmission test on the radio frequency transceiver to generate a plurality of circuit test results so as to select a target simulation circuit from a plurality of reference simulation circuits, and to match the radio frequency transceiver according to the target simulation circuit. The target simulation circuit corresponds to a lowest value among the plurality of circuit test results.

(6) According to another aspect of the present disclosure, a calibration system for a radio frequency transceiver includes a radio frequency transceiver, a memory unit, and a controller. The memory unit is for storing a first calibration software module, a second calibration software module, a circuit matching software module, a plurality of reference calibration models, a plurality of first parameters, a plurality of second parameters, and a plurality of reference simulation circuits. The controller is connected to the memory unit and the radio frequency transceiver and configured to perform a first calibration step, a second calibration step, and a circuit matching step. In the first calibration step, the controller executes the first calibration software module for performing a first transmission test on the radio frequency transceiver to generate a plurality of model test results, selects a target calibration model from the plurality of reference calibration models, and calibrates the radio frequency transceiver according to the target calibration model. The target calibration model corresponds to a lowest value among the plurality of model test results. In the second calibration step, the controller executes the second calibration software module for performing a second transmission test on the radio frequency transceiver to generate a plurality of first test results and a plurality of second test results, selects a first target parameter from the plurality of first parameters and a second target parameter from the plurality of second parameters, and calibrates the radio frequency transceiver according to the first target parameter and the second target parameter. The first target parameter corresponds to a lowest value among the plurality of first test results, and the second target parameter corresponds to a lowest value among the plurality of second test results. In the circuit matching step, the controller executes the circuit matching software module for performing a third transmission test on the radio frequency transceiver to generate a plurality of circuit test results, selects a target simulation circuit from the plurality of reference simulation circuits, and matches the target simulation circuit to the radio frequency transceiver. The target simulation circuit corresponds to a lowest value among the plurality of circuit test results.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:
- (2) FIG. 1 is a block diagram of a calibration system for a radio frequency transceiver according to a first embodiment of the present disclosure.
- (3) FIG. 2 is a flow chart of a calibration method for a radio frequency transceiver according to a second embodiment of the present disclosure.
- (4) FIG. 3 is a schematic diagram illustrating a test environment of the calibration method for the radio frequency transceiver shown in FIG. 2.
- (5) FIG. 4 is a flow chart of a first calibration step in the calibration method for the radio frequency transceiver shown in FIG. 2.
- (6) FIG. 5 is a flow chart of a second calibration step in the calibration method for the radio frequency transceiver shown in FIG. 2.
- (7) FIG. 6 is a flow chart of a circuit matching step in the calibration method for the radio frequency transceiver shown in FIG. 2.
- (8) FIG. 7 is a schematic diagram of an impedance matching module in the radio frequency transceiver shown in FIG. 3.
- (9) FIG. 8 is a flow chart of a voltage adjustment step in the calibration method for the radio frequency transceiver shown in FIG. 2.

DETAILED DESCRIPTION

- (10) The embodiment will be described with the drawings. For clarity, some practical details will be described below. However, it should be noted that the present disclosure should not be limited by the practical details, that is, in some embodiment, the practical details is unnecessary. In addition, for simplifying the drawings, some conventional structures and elements will be simply illustrated, and repeated elements can be represented by the same labels.
- (11) It will be understood that when an element (or device) is referred to as be “connected to” another element, it can be directly connected to the other element, or it can be indirectly connected to the other element, that is, intervening elements can be present. In contrast, when an element is referred to as be “directly connected to” another element, there are no intervening elements present. In addition, the terms first, second, third, etc. are used herein to describe various elements or components, these elements or components should not be limited by these terms. Consequently, a first element or component discussed below could be termed a second element or component.
- (12) FIG. 1 is a block diagram of a calibration system **100** for a radio frequency transceiver (hereinafter referred to as “the calibration system **100**”) according to a first embodiment of the present disclosure. Referring to FIG. 1, the calibration system **100** includes a radio frequency transceiver **200**, a memory unit **300**, a controller **400**, and a test equipment **500**.
- (13) The radio frequency transceiver **200** can be, but not limited to a remote radio unit (RRU) used in open radio access network (O-RAN) or a communication device that operates in the 2.4G/5G frequency band.
- (14) The memory unit **300** is a machine-readable medium, which can be, but is not limited to a random access memory (RAM), a read-only memory (ROM), a flash memory, a hard disk drive, or a combination thereof, and stores a first calibration software module **310**, a second calibration software module **320**, a circuit matching software module **330**, a voltage adjustment software module **340**, and a database **350**. Each of the aforementioned software modules can be programmed by any one or more programming languages, and can include, but not limited to object-oriented programming languages (such as Java, C++, C#), general-purpose programming languages (such as C language, Visual Basic), dynamic programming languages (such as Python), or other

programming languages. The database **350** stores a plurality of reference calibration models, a plurality of reference parameters, a plurality of reference simulation circuits, and a plurality of reference gate source voltages.

(15) The controller **400** is signally connected to the memory unit **300** and the radio frequency transceiver **200** and is configured to implement a calibration method for a radio frequency transceiver, which includes a first calibration step, a second calibration step, a circuit matching step and a voltage adjustment step. The controller **400** calibrates the radio frequency transceiver **200** through the aforementioned steps, so that the radio frequency signal transmitted by the radio frequency transceiver **200** is linearized. The controller **400** can be a digital signal processor (DSP), a micro processing unit (MPU), a central processing unit (CPU) or other electronic processors. The controller **400** executes the first calibration software module **310**, the second calibration software module **320**, the circuit matching software module **330** and the voltage adjustment software module **340** respectively to perform the first, second, third and fourth transmission tests on the radio frequency transceiver **200**, and reads the data stored in the database **350** from the memory unit **300** to implement the calibration method for the radio frequency transceiver according to the present disclosure.

(16) The test equipment **500** is signally connected to the radio frequency transceiver **200** and the controller **400**, and can be a wireless test instrument (such as Keysight E6680A) or a calibration device. The test equipment **500** is used to perform multiplex testing on the radio frequency signals transmitted by the radio frequency transceiver **200** to generate test results, and transmit multiple test results to the controller **400**. The controller **400** selects the target calibration model, the target parameter set, the target simulation circuit, and the target gate source voltage from the database **350** based on the test results, and implements and adjusts the internal parameter settings and circuit configuration of the radio frequency transceiver **200** accordingly. Thus, manual testing man-hours can be effectively reduced and the performance of the radio frequency transceiver **200** can be improved. In one embodiment, the controller of the present disclosure can be signally connected to a cloud server, which can also store the aforementioned software modules and database. In another embodiment, the controller and memory unit can be internal components of a smart device and/or computer device. The detailed operation of each step of the calibration method for the radio frequency transceiver according to the present disclosure will be described hereafter in conjunction with the figures.

(17) FIG. 2 is a flow chart of a calibration method **S0** for a radio frequency transceiver (hereinafter referred to as “the calibration method **S0**”) according to a second embodiment of the present disclosure. Referring to FIG. 1 and FIG. 2, the calibration method **S0** includes a first calibration step **S02**, a second calibration step **S04**, a circuit matching step **S06**, and a voltage adjustment step **S08**, and is implemented by the calibration system **100** of the first embodiment. The calibration method **S0** uses the controller **400** to execute the first calibration step **S02**, the second calibration step **S04**, the circuit matching step **S06** and the voltage adjustment step **S08** in sequence to calibrate the radio frequency transceiver **200**, so that the radio frequency signals transmitted by the radio frequency transceiver **200** are linearized.

(18) The first calibration step **S02** includes configuring the controller **400** to execute the first calibration software module **310** to perform a first transmission test on the radio frequency transceiver **200** and generate a plurality of model test results so as to select the target calibration model from the plurality of reference calibration models stored in the database **350** of the memory unit **300**. The radio frequency transceiver **200** is then calibrated according to the target calibration model, wherein the target calibration model corresponds to a lowest value among the plurality of model test results.

(19) The second calibration step **S04** includes configuring the controller **400** to execute the second calibration software module **320** to perform a second transmission test on the radio frequency transceiver **200** and generate a plurality of parameter test results so as to select the target parameter

set from a plurality of reference parameters stored in the database **350** of the memory unit **300**. The radio frequency transceiver **200** is then calibrated according to the target parameter set, wherein the target parameter set corresponds to a value set of the plurality of parameter test results. In specific, the plurality of parameter test results can include a plurality of first test results and a plurality of second test results, that is, some of the parameter test results are the first test results, and the remaining parameter test results are the second test results. The plurality of reference parameters can include a plurality of first parameters and a plurality of second parameters, that is, some of the reference parameters are the first parameters, and the remaining reference parameters are the second parameters. In the second calibration step **S04**, the controller **400** selects a first target parameter from the first parameters, and selects a second target parameter from the second parameters. The first target parameter and the second target parameter form the target parameter set. The first target parameter corresponds to the lowest value among the first test results, and the second target parameter corresponds to the lowest value among the second test results. The lowest value among the first test results and the lowest value among the second test results form the value set.

(20) The circuit matching step **S06** includes configuring the controller **400** to execute the circuit matching software module **330** to perform a third transmission test on the radio frequency transceiver **200** and generate a plurality of circuit test results so as to select the target simulation circuit from the plurality of reference simulation circuits stored in the database **350** of the memory unit **300**. The target simulation circuit is then matched to the radio frequency transceiver **200**, wherein the target simulation circuit corresponds to a lowest value among the plurality of circuit test results.

(21) The voltage adjustment step **S08** includes configuring the controller **400** to execute the voltage adjustment software module **340** to perform a fourth transmission test on the radio frequency transceiver **200** and generate a plurality of voltage test results so as to select the target gate source voltage from a plurality of reference gate source voltages stored in the database **350** of the memory unit **300**. The target gate source voltage is then applied to the radio frequency transceiver **200**, wherein the target gate source voltage corresponds to a lowest value among the plurality of voltage test results.

(22) Hence, the calibration method **S0** of the present disclosure uses the controller **400** to automatically and sequentially perform the first, second, third and fourth transmission tests on the radio frequency transceiver **200**, and respective selects, from the first, second, third and fourth transmission tests, the target calibration model, target parameter set, target simulation circuit and target gate source voltage that correspond to the lowest value among the test results, and then implements and adjusts the internal parameter setting and circuit configuration of the radio frequency transceiver **200** accordingly, thereby effectively reducing manual testing hours and improving the performance of the radio frequency transceiver **200**.

(23) FIG. **3** is a schematic diagram illustrating a test environment of the calibration method **S0** shown in FIG. **2**, and FIG. **4** is a flow chart of a first calibration step **S02** in the calibration method **S0** shown in FIG. **2**. Referring to FIG. **1** to FIG. **4**, the radio frequency transceiver **200** includes a baseband module **210**, a radio frequency module **220**, a power amplifier module **230**, an impedance matching module **240**, and an antenna module **250**. The radio frequency module **220** is electrically connected to the baseband module **210**, the power amplifier module **230** is electrically connected to the radio frequency module **220**, the impedance matching module **240** is electrically connected to the power amplifier module **230**, and the antenna module **250** is electrically connected to the impedance matching module **240**. The first calibration step **S02** further includes a radio frequency signal generation step **S022**, a signal checking step **S024**, and a model selection step **S026**.

(24) The radio frequency signal generation step **S022** is to configure the controller **400** to transmit the model calibration command **401** to the baseband module **210** during the first transmission test, and control the baseband module **210** through the model calibration command **401** to drive the

radio frequency module **220** to generate a plurality of radio frequency signals **201** according to the plurality of reference calibration models. More specifically, after the model calibration command **401** generated by the controller **400** is processed by the baseband module **210**, the baseband module **210** transmits a baseband signal to the radio frequency module **220**. Then, the radio frequency module **220** converts the baseband signal into an initial radio frequency signal. The power amplifier module **230** is used to amplify the output power of the initial radio frequency signal and convert the initial radio frequency signal into the radio frequency signal **201**. The impedance matching module **240** is used to achieve impedance matching between the power amplifier module **230** and the antenna module **250**. Finally, the antenna module **250** is used to transmit the radio frequency signal **201** to the test equipment **500**.

(25) The signal checking step **S024** includes configuring the test equipment **500** to receive and check the radio frequency signals **201** to generate a plurality of model test results **501** corresponding to the plurality of reference calibration models. The model selection step **S026** includes configuring the controller **400** to select the lowest value among the plurality of model test results **501**, and determine that the reference calibration model corresponding to the lowest value among the plurality of model test results **501** is the target calibration model. Specifically, each reference calibration model stored in the database **350** can be a digital pre-distortion (DPD) calibration model or a combination of DPD calibration parameters. When transmitting the model calibration command **401** to the baseband module **210**, the controller **400** will also write different reference calibration models into the radio frequency module **220**. Therefore, in the first transmission test, the radio frequency module **220** generates corresponding radio frequency signals **201** according to different reference calibration models. The test equipment **500** generates corresponding model test results **501** according to different radio frequency signals **201**, and each model test result **501** can correspond to a value (%) of an error vector magnitude (EVM).

(26) After the first transmission test is completed, the controller **400** selects the reference calibration model with the lowest EVM value as the target calibration model according to the plurality of model test results **501**, and then calibrates the radio frequency transceiver **200** according to the target calibration model. In addition, the controller **400** sorts the model test results **501** in a numerical order based on the EVM values to generate a test result list **360**, and stores the test result list **360** in the memory unit **300**. In other words, the controller **400** records the reference calibration models and their corresponding model test results **501** in the memory unit **300** in a comma-separated value file format (such as, but not limited to, file extensions csv, txt or other file formats), so the user can review the test result list **360** and confirm whether the selected target calibration model is the best among all reference calibration models.

(27) FIG. 5 is a flow chart of the second calibration step **S04** in the calibration method **S0** shown in FIG. 2. Referring to FIG. 1, FIG. 2, FIG. 3, and FIG. 5, the second calibration step **S04** further includes a radio frequency signal generation step **S042**, a signal checking step **S044**, a first parameter selection step **S046** and a second parameter selection step **S048**.

(28) The radio frequency signal generation step **S042** includes configuring the controller **400** to transmit the parameter calibration command **402** to the baseband module **210** in the second transmission test, and control the baseband module **210** through the parameter calibration command **402** to drive the radio frequency module **220** to respectively generate a plurality of first radio frequency signals **202a** and a plurality of second radio frequency signals **202b** according to the plurality of first parameters and the plurality of second parameters.

(29) The signal checking step **S044** includes configuring the test equipment **500** to receive and check the first radio frequency signals **202a** and the second radio frequency signals **202b** to respectively generate a plurality of first test results **502a** corresponding to the plurality of first parameters and a plurality of second test results **502b** corresponding to the plurality of second parameters.

(30) The first parameter selection step **S046** includes configuring the controller **400** to select the

lowest value among the first test results **502a**, and determine that the first parameter corresponding to the lowest value among the first test results **502a** is the first target parameter. The second parameter selection step **S048** includes configuring the controller **400** to select the lowest value among the second test results **502b**, and determine that the second parameter corresponding to the lowest value among the second test results **502b** is the second target parameter. In particular, each first parameter stored in the database **350** can be a regularization parameter, and each second parameter stored in the database **350** can be a damping factor. The radio frequency module **220** generates corresponding first radio frequency signals **202a** and second radio frequency signals **202b** according to different first parameters and second parameters, respectively. The test equipment **500** generates corresponding first test results **502a** and second test results **502b** respectively according to different first radio frequency signals **202a** and second radio frequency signals **202b**.

(31) Moreover, the first test results **502a** and the second test results **502b** each correspond to a value (dB) of an adjacent channel leakage power ratio (ACLR). After the second transmission test is completed, the controller **400** selects the first parameter with the lowest ACLR value as the first target parameter according to the plurality of first test results **502a**, and selects the second parameter with the lowest ACLR value as the second target parameter according to the plurality of second test results **502b**. Thus, the controller **400** can calibrate the radio frequency transceiver **200** according to the first target parameter and the second target parameter, so that the power leakage of the radio frequency transceiver **200** in adjacent channels is controlled below a specific level, thereby reducing the interference to the radio frequency transceiver **200**.

(32) FIG. **6** is a flow chart of the circuit matching step **S06** in the calibration method **S0** shown in FIG. **2**, and FIG. **7** is a schematic diagram of the impedance matching module **240** in the radio frequency transceiver **200** shown in FIG. **3**. Referring to FIG. **1**, FIG. **2**, FIG. **3**, FIG. **6**, and FIG. **7**, the circuit matching step **S06** further includes an impedance connection step **S062**, a radio frequency signal generating step **S064**, a signal checking step **S066** and a circuit selection step **S068**.

(33) The impedance connection step **S062** is to electrically connect the impedance matching module **240** between the output end of the power amplifier module **230** and the input end of the antenna module **250**. The impedance matching module **240** includes an antenna switch **241**, which can be an SP5T switch and has a plurality of paths **P1**, **P2**, **P3**, **P4**, and **P5**, and the paths **P1**, **P2**, **P3**, **P4**, and **P5** are respectively connected to a default simulation circuit **C0** and at least one reference simulation circuits **C1**, **C2**, **C3**, **C4**. In other embodiments, the antenna switch can be, but is not limited to, one or more multi-throw switches (such as SP10T). The configuration of the antenna switch is determined by the user's needs and the number of reference simulation circuits in the database.

(34) The radio frequency signal generation step **S064** is to configure the controller **400** to selectively switch the paths **P1**, **P2**, **P3**, **P4**, and **P5** of the antenna switch **241** of the impedance matching module **240** in the third transmission test to connect respectively to the default simulation circuits **C0** and the reference simulation circuits **C1**, **C2**, **C3**, **C4**, and to transmit a circuit calibration command **403** to the baseband module **210** to control the baseband module **210** through the circuit calibration command **403** to drive the radio frequency module **220** to generate a plurality of radio frequency signals **203** according to the default simulation circuit **C0** and the reference simulation circuits **C1**, **C2**, **C3**, and **C4**.

(35) The signal checking step **S066** is to configure the test equipment **500** to check the radio frequency signals **203** to generate a circuit test result **503** corresponding to the default simulation circuit **C0** and a plurality of circuit test results **503** corresponding to the reference simulation circuits **C1**, **C2**, **C3**, and **C4**.

(36) The circuit selection step **S068** is to configure the controller **400** to select the lowest value among the circuit test results **503**, and determine one of the default simulation circuit **C0** and the

reference simulation circuits C1, C2, C3, and C4 that corresponds to the lowest value among the circuit test results 503 to be the target simulation circuit.

(37) Specifically, in both the first calibration step S02 and the second calibration step S04, the controller 400 switches the antenna switch 241 to the path P1 to connect to the default simulation circuit C0, so that the impedance matching module 240 applies the same impedance value in the first transmission test and the second transmission test, for example 0 ohm. In the circuit matching step S06, each of the reference simulation circuits C1, C2, C3, and C4 stored in the database 350 has different impedance values. For example, the reference simulation circuits C1, C2, C3, and C4 can respectively correspond to any coordinate point in the four quadrants of the Smith chart. The radio frequency module 220 generates corresponding radio frequency signals 203 according to different impedance values (i.e., the default simulation circuit C0 and the reference simulation circuits C1, C2, C3, and C4). The test equipment 500 generates corresponding circuit test results 503 according to different radio frequency signals 203, wherein each circuit test result 503 can correspond to a value (dB) of ACLR. The controller 400 selects one of the default simulation circuit C0 and the reference simulation circuits C1, C2, C3, C4 with the lowest ACLR value as the target simulation circuit according to the plurality of circuit test results 503. Thereby, the controller 400 can match the target simulation circuit to the radio frequency transceiver 200, which can not only reduce the interference received by the radio frequency transceiver 200, but also enable the radio frequency transceiver 200 to achieve impedance matching.

(38) FIG. 8 is a flow chart of the voltage adjustment step S08 in the calibration method S0 shown in FIG. 2. Referring to FIG. 1, FIG. 2, FIG. 3 and FIG. 8, the voltage adjustment step S08 further includes a radio frequency signal generation step S082, a signal checking step S084, and a voltage selection step S086.

(39) The radio frequency signal generation step S082 is to configure the controller 400 to respectively apply the plurality of reference gate source voltages in the database 350 to the power amplifier module 230 in the fourth transmission test, and transmit a voltage calibration command 404 to the baseband module 210. The controller 400 controls the baseband module 210 through the voltage calibration command 404 to drive the radio frequency module 220 to generate a plurality of radio frequency signals 204 according to the reference gate source voltages (that is, the radio frequency module 220 generates the corresponding radio frequency signals 204 according to the different reference gate source voltages applied to the power amplifier module 230).

(40) The signal checking step S084 is to configure the test equipment 500 to receive and check the radio frequency signals 204 to generate a plurality of voltage test results 504 corresponding to the plurality of reference gate source voltages.

(41) The voltage selection step S086 is to configure the controller 400 to select the lowest value among the voltage test results 504, and determine that the reference gate source voltage corresponding to the lowest value among the voltage test results 504 is the target gate source voltage, wherein each voltage test result 504 corresponds to a value of ACLR (dB). After the fourth transmission test is completed, the controller 400 selects the reference gate source voltage with the lowest ACLR value as the target gate source voltage according to the plurality of voltage test results 504. Thereby, the controller 400 can further optimize the radio frequency transceiver 200 by applying the target gate source voltage to the power amplifier module 230 of the radio frequency transceiver 200.

(42) Furthermore, the voltage adjustment step S08 of the calibration method S0 can also be replaced by a current adjustment step, which means that the aforementioned fourth transmission test can be changed from a voltage test to a current test, and the database 350 of the memory unit 300 further stores a plurality of reference current parameters. The current adjustment step includes configuring the controller 400 to execute a current adjustment software module to perform another fourth transmission test on the radio frequency transceiver 200 to generate a plurality of current test results and select a target current parameter from the plurality of reference current parameters so as

to apply the target current parameter to the radio frequency transceiver **200**, wherein the target current parameter corresponds to a lowest value among one of the current test results. In addition, the current adjustment step can further include a radio frequency signal generation step, a signal checking step and a current selection step.

(43) The aforementioned radio frequency signal generation step is to configure the controller **400** to respectively apply the plurality of reference current parameters in the database **350** to the power amplifier module **230** in the another fourth transmission test, and transmit a current calibration command to the baseband module **210**, and then the baseband module **210** is controlled by the current calibration command to drive the radio frequency module **220** to generate a plurality of radio frequency signals according to these reference current parameters. The aforementioned signal checking step is to configure the test equipment **500** to receive and check these radio frequency signals to generate a plurality of current test results corresponding to the plurality of reference current parameters. The aforementioned current selection step is to configure the controller **400** to select the lowest value among these current test results, and determine that the reference current parameter corresponding to the lowest value among these current test results is the target current parameter, wherein each current test result corresponds to an ACLR value (dB). After the other fourth transmission test is completed, the controller can select the reference current parameter with the lowest ACLR value as the target current parameter according to the plurality of current test results. Thereby, the controller **400** can further optimize the radio frequency transceiver **200** not only by applying the target gate source voltage to the power amplifier module **230**, but also by applying the target current parameter to the power amplifier module **230**.

(44) To sum up, the present disclosure has the following advantages. First, the controller is used to automatically and sequentially conduct multiple transmission tests on the radio frequency transceiver, and gradually select the target calibration model, target parameter set, target simulation circuit and target gate source voltage (or target current parameter), and then implement it and adjust the parameter settings and circuit configuration of the radio frequency transceiver to optimize the radio frequency transceiver. Secondly, it can effectively reduce manual testing man-hours and improve the performance of radio frequency transceivers, and solve the quality issue (performance not as good as expected) of the sold products due to human error in testing.

(45) Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

(46) It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the present disclosure cover modifications and variations of this disclosure provided they fall within the scope of the following claims.

Claims

1. A calibration method for a radio frequency transceiver, the calibration method comprising: a first calibration step comprising configuring a controller to execute a first calibration software module for performing a first transmission test on the radio frequency transceiver to generate a plurality of model test results so as to select a target calibration model from a plurality of reference calibration models, and to calibrate the radio frequency transceiver according to the target calibration model, wherein the target calibration model corresponds to a lowest value among the plurality of model test results; a second calibration step comprising configuring the controller to execute a second calibration software module for performing a second transmission test on the radio frequency transceiver to generate a plurality of parameter test results so as to select a target parameter set from a plurality of reference parameters, and to calibrate the radio frequency transceiver according to the target parameter set, wherein the target parameter set corresponds to a value set of the

plurality of parameter test results; and a circuit matching step comprising configuring the controller to execute a circuit matching software module for performing a third transmission test on the radio frequency transceiver to generate a plurality of circuit test results so as to select a target simulation circuit from a plurality of reference simulation circuits, and to match the radio frequency transceiver according to the target simulation circuit, wherein the target simulation circuit corresponds to a lowest value among the plurality of circuit test results.

2. The calibration method of claim 1, wherein each of the plurality of reference calibration models is a digital pre-distortion (DPD) calibration model, some of the plurality of reference parameters are respectively regularization parameters, and remaining of the plurality of reference parameters are respectively damping factors.

3. The calibration method of claim 1, wherein the radio frequency transceiver comprises a baseband module and a radio frequency module, and the first calibration step further comprises: a radio frequency signal generation step comprising configuring the controller to transmit a model calibration command to the baseband module during the first transmission test for controlling the baseband module to drive the radio frequency module to generate a plurality of radio frequency signals according to the plurality of reference calibration models; a signal checking step comprising configuring a test equipment to check the plurality of radio frequency signals and generate the plurality of model test results corresponding to the plurality of reference calibration models; and a model selection step comprising configuring the controller to select the lowest value among the plurality of model test results and determine one of the plurality of reference calibration models that corresponds to the lowest value among the plurality of model test results to be the target calibration model.

4. The calibration method of claim 3, wherein each of the plurality of model test results corresponds to an error vector magnitude (EVM) value.

5. The calibration method of claim 1, wherein the controller generates a test result list by sorting the plurality of model test results in numerical order after the first transmission test and stores the test result list in a memory unit.

6. The calibration method of claim 1, wherein the plurality of parameter test results comprise a plurality of first test results and a plurality of second test results, the plurality of reference parameters comprise a plurality of first parameters and a plurality of second parameters; and the controller selects a first target parameter from the plurality of first parameters and a second target parameter from the plurality of second parameters, the first target parameter and the second target parameter form the target parameter set, the first target parameter corresponds to a lowest value among the plurality of first test results, the second target parameter corresponds to a lowest value among the plurality of second test results, and the lowest value among the plurality of first test results and the lowest value among the plurality of second test results form the value set.

7. The calibration method of claim 6, wherein the radio frequency transceiver comprises a baseband module and a radio frequency module, and the second calibration step further comprises: a radio frequency signal generation step comprising configuring the controller to transmit a parameter calibration command to the baseband module during the second transmission test for controlling the baseband module to drive the radio frequency module to respectively generate a plurality of first radio frequency signals and a plurality of second radio frequency signals according to the plurality of first parameters and the plurality of second parameters; a signal checking step comprising configuring a test equipment to check the plurality of first radio frequency signals and the plurality of second radio frequency signals and generate the plurality of first test results corresponding to the plurality of first parameters and the plurality of second test results corresponding to the plurality of second parameters; a first parameter selection step comprising configuring the controller to select the lowest value among the plurality of first test results and determine one of the plurality of first parameters that corresponds to the lowest value among the plurality of first test results to be the first target parameter; and a second parameter selection step

comprising configuring the controller to select the lowest value among the plurality of second test results and determine one of the plurality of second parameters that corresponds to the lowest value among the plurality of second test results to be the second target parameter.

8. The calibration method of claim 6, wherein each of the plurality of first test results and each of the plurality of second test results correspond respectively to an adjacent channel leakage power ratio (ACLR) value.

9. The calibration method of claim 1, wherein the radio frequency transceiver comprises a baseband module, a radio frequency module, and a power amplifier module, and the circuit matching step further comprises: an impedance connection step comprising connecting an impedance matching module to an output end of the power amplifier module; a radio frequency signal generation step comprising configuring the controller to selectively switch a plurality of paths of the impedance matching module during the third transmission test to respectively connect the plurality of reference simulation circuits and transmit a circuit calibration command to the baseband module for controlling the baseband module to drive the radio frequency module to generate a plurality of radio frequency signals according to the plurality of reference simulation circuits; a signal checking step comprising configuring a test equipment to check the plurality of radio frequency signals and generate the plurality of circuit test results corresponding to the plurality of reference simulation circuits; and a circuit selection step comprising configuring the controller to select the lowest value among the plurality of circuit test results and determine one of the plurality of reference simulation circuits that corresponds to the lowest value among the plurality of circuit test results to be the target simulation circuit.

10. The calibration method of claim 9, wherein each of the plurality of circuit test results corresponds to an adjacent channel leakage power ratio (ACLR) value.

11. The calibration method of claim 1, further comprising: a voltage adjustment step comprising configuring the controller to execute a voltage adjustment software module for performing a fourth transmission test on the radio frequency transceiver to generate a plurality of voltage test results so as to select a target gate source voltage from a plurality of reference gate source voltages, and apply the target gate source voltage to the radio frequency transceiver, wherein the target gate source voltage corresponds to a lowest value among the plurality of voltage test results; wherein the controller executes the first calibration step, the second calibration step, the circuit matching step, and the voltage adjustment step in sequence.

12. The calibration method of claim 11, wherein the radio frequency transceiver comprises a baseband module, a radio frequency module, and a power amplifier module, and the voltage adjustment step further comprises: a radio frequency signal generation step comprising configuring the controller to apply the plurality of reference gate source voltages to the power amplifier module respectively during the fourth transmission test and transmit a voltage calibration command to the baseband module for controlling the baseband module to drive the radio frequency module to generate a plurality of radio frequency signals according to the plurality of reference gate source voltages; a signal checking step comprising configuring a test equipment to check the plurality of radio frequency signals and generate the plurality of voltage test results corresponding to the plurality of reference gate source voltages; and a voltage selection step comprising configuring the controller to select the lowest value among the plurality of voltage test results and determine one of the plurality of reference gate source voltages that corresponds to the lowest value among the plurality of voltage test results to be the target gate source voltage.

13. A calibration system for a radio frequency transceiver, the calibration system comprising: a radio frequency transceiver; a memory unit for storing a first calibration software module, a second calibration software module, a circuit matching software module, a plurality of reference calibration models, a plurality of first parameters, a plurality of second parameters, and a plurality of reference simulation circuits; and a controller connected to the memory unit and the radio frequency transceiver and configured to perform a first calibration step, a second calibration step,

and a circuit matching step; wherein, in the first calibration step, the controller executes the first calibration software module for performing a first transmission test on the radio frequency transceiver to generate a plurality of model test results, selects a target calibration model from the plurality of reference calibration models, and calibrates the radio frequency transceiver according to the target calibration model, wherein the target calibration model corresponds to a lowest value among the plurality of model test results; wherein, in the second calibration step, the controller executes the second calibration software module for performing a second transmission test on the radio frequency transceiver to generate a plurality of first test results and a plurality of second test results, selects a first target parameter from the plurality of first parameters and a second target parameter from the plurality of second parameters, and calibrates the radio frequency transceiver according to the first target parameter and the second target parameter, wherein the first target parameter corresponds to a lowest value among the plurality of first test results, and the second target parameter corresponds to a lowest value among the plurality of second test results; and wherein, in the circuit matching step, the controller executes the circuit matching software module for performing a third transmission test on the radio frequency transceiver to generate a plurality of circuit test results, selects a target simulation circuit from the plurality of reference simulation circuits, and matches the target simulation circuit to the radio frequency transceiver, wherein the target simulation circuit corresponds to a lowest value among the plurality of circuit test results.

14. The calibration system of claim 13, wherein each of the plurality of reference calibration models is a digital pre-distortion (DPD) calibration model, each of the plurality of first parameters is a regularization parameter, and each of the plurality of second parameters is a damping factor.

15. The calibration system of claim 13, further comprising: a test equipment connected to the radio frequency transceiver and the controller; wherein the radio frequency transceiver comprises a baseband module and a radio frequency (RF) module connected to the baseband module, and the first calibration step further comprises a radio frequency signal generation step, a signal checking step, and a model selection step; wherein, in the radio frequency signal generation step, the controller transmits a model calibration command to the baseband module during the first transmission test, controls the baseband module through the model calibration command to drive the radio frequency module to generate a plurality of radio frequency signals according to the plurality of reference calibration models; wherein, in the signal checking step, the test equipment checks the plurality of radio frequency signals and generates the plurality of model test results corresponding to the plurality of reference calibration models; and wherein, in the model selection step, the controller selects the lowest value among the plurality of model test results and determines one of the plurality of reference calibration models that corresponds to the lowest value among the plurality of model test results to be the target calibration model; wherein, each of the plurality of model test results corresponds to an error vector magnitude (EVM) value.

16. The calibration system of claim 13, wherein the controller generates a test result list by sorting the plurality of model test results in numerical order after the first transmission test and stores the test result list in the memory unit.

17. The calibration system of claim 13, further comprising: a test equipment connected to the radio frequency transceiver and the controller; wherein the radio frequency transceiver comprises a baseband module and a radio frequency module connected to the baseband module, and the second calibration step further comprises a radio frequency signal generation step, a signal checking step, a first parameter selection step, and a second parameter selection step; wherein, in the radio frequency signal generation step, the controller transmits a parameter calibration command to the baseband module during the second transmission test and controls the baseband module through the parameter calibration command to drive the radio frequency module to respectively generate a plurality of first radio frequency signals and a plurality of second radio frequency signals according to the plurality of first parameters and the plurality of second parameters; wherein, in the signal checking step, the test equipment checks the plurality of first radio frequency signals and the

plurality of second radio frequency signals and respectively generates the plurality of first test results corresponding to the plurality of first parameters and the plurality of second test results corresponding to the plurality of second parameters; wherein, in the first parameter selection step, the controller selects the lowest value among the plurality of first test results and determines the one of the plurality of first parameters that corresponds to the lowest value among the plurality of first test results to be the first target parameter; and wherein, in the second parameter selection step, the controller selects the lowest value among the plurality of second test results and determines the one of the plurality of second parameters that corresponds to the lowest value among the plurality of second test results to be the second target parameter; wherein each of the plurality of first test results corresponds to an adjacent channel leakage power ratio (ACLR) value, and each of the plurality of second test results corresponds to an adjacent channel leakage power ratio (ACLR) value.

18. The calibration system of claim 13, further comprising: a test equipment connected to the radio frequency transceiver and the controller; wherein the radio frequency transceiver comprises: a baseband module; a radio frequency module connected to the baseband module; a power amplifier module connected to the radio frequency module; and an impedance matching module connected to the power amplifier module and comprising a plurality of paths; wherein the circuit matching step further comprises a radio frequency signal generation step, a signal checking step, and a circuit selection step; wherein, in the radio frequency signal generation step, the controller selectively switches the plurality of paths in the impedance matching module to connect respectively to the plurality of reference simulation circuits during the third transmission test, transmits a circuit calibration command to the baseband module, and controls the baseband module through the circuit calibration command to drive the radio frequency module to generate a plurality of radio frequency signals according to the plurality of reference simulation circuits; wherein, in the signal checking step, the test equipment checks the plurality of radio frequency signals and generates the plurality of circuit test results corresponding to the plurality of reference simulation circuits; and wherein, in the circuit selection step, the controller selects the lowest value among the plurality of circuit test results and determines one of the plurality of reference simulation circuits that corresponds to the lowest value among the plurality of circuit test results to be the target simulation circuit; wherein each of the plurality of circuit test results corresponds to an adjacent channel leakage power ratio (ACLR) value.

19. The calibration system of claim 13, wherein the controller is configured to perform a voltage adjustment step, and in the voltage adjustment step, the controller executes a voltage adjustment software module for performing a fourth transmission test on the radio frequency transceiver to generate a plurality of voltage test results, selects a target gate source voltage from a plurality of reference gate source voltages, and applies the target gate source voltage to the radio frequency transceiver, and the target gate source voltage corresponds to a lowest value among the plurality of voltage test results.

20. The calibration system of claim 19, further comprising: a test equipment connected to the radio frequency transceiver and the controller; wherein the radio frequency transceiver comprises: a baseband module; a radio frequency module connected to the baseband module; and a power amplifier module connected to the radio frequency module; wherein the voltage adjustment step further comprises a radio frequency signal generation step, a signal checking step, and a voltage selection step; wherein, in the radio frequency signal generation step, the controller apply the plurality of reference gate source voltages to the power amplifier module respectively during the fourth transmission test, transmits a voltage calibration command to the baseband module, and controls the baseband module through the voltage calibration command to drive the radio frequency module to generate a plurality of radio frequency signals according to the plurality of reference gate source voltages; wherein, in the signal checking step, the test equipment checks the plurality of radio frequency signals and generates the plurality of voltage test results corresponding

to the plurality of reference gate source voltages; and wherein, in the voltage selection step, the controller selects the lowest value among the plurality of voltage test results and determines one of the plurality of reference gate source voltages that corresponds to the lowest value among the plurality of voltage test results to be the target gate source voltage.
